# Systemadministration

Linux

## DHCP

# Agenda

1. DHCP
2. DHCP Relay
3. DHCP Redundancy
4. DHCP Options
5. DHCP Leases
6. ISC Kea
7. DHCP Troubleshooting

# Dynamic Host Configuration Protocol

# **D**ynamic **H**ost **C**onfiguration **P**rotocol

1984 1985 1993

**RARP → BOOTP → DHCP**

**DHCP** dynamisk konfiguration av:

| | |
|---|---|
| IP-adress | Hostname |
| Nätmask | NTP-servrar |
| Default gateway | …osv |
| DNS-servrar | |

# DHCP-begrepp

**Scope**

- Det spann av adresser som kan delas ut

**Lease**

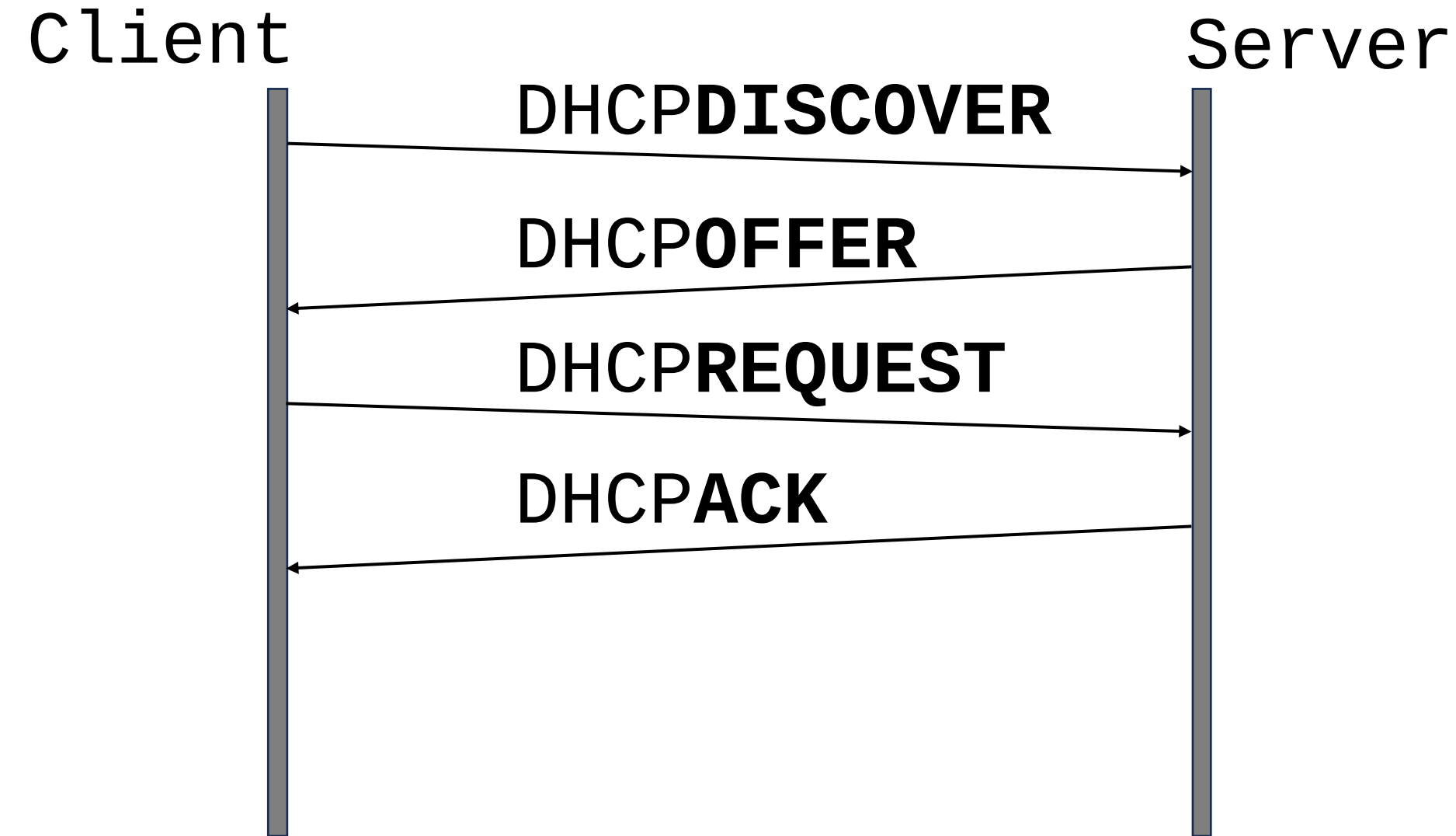
- En utdelad adress med inställningar

**Options**

- Inställningar som kan delas ut med DHCP

*Implementation example of Peg DHCP (RFC 2322): Pegs and network information used on What the Hack 2005*, by Ministry of Truth, 2006, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Peg_dhcp_from_what_the_hack_holland_2005.jpg). CC-BY-SA-2.5.

# DHCP Messages



# DHCP Message Format

| 8 | 16 | 24 | 32 |
|---|---|---|---|
| OP Code (1) | Hardware Type (1) | Hardware Address Length (1) | Hops (1) |
| Transaction Identifier | | | |
| Seconds - 2 bytes | | Flags - 2 bytes | |
| Client IP Address (CIADDR) - 4 bytes | | | |
| Your IP Address (YIADDR) - 4 bytes | | | |
| Server IP Address (SIADDR) - 4 bytes | | | |
| Gateway IP Address (GIADDR) - 4 bytes | | | |
| Client Hardware Address (CHADDR) - 16 bytes | | | |
| Server Name (SNAME) - 64 bytes | | | |
| Boot Filename - 128 bytes | | | |
| DHCP Options - variable | | | |

DHCP Message Type

Klientens nuvarande adress

Serverns adress

Options

Erbjudna adress till klienten

Relay address

# DHCP Messages Example

Client

- MAC address: 70:5A:0F:38:C0:F9
- IP address: 0.0.0.0 → 192.168.1.10

Server

- MAC address: 00:08:E3:FF:FD:90
- IP address: 192.168.1.5

Anta: Första gången klienten ansluter, det finns ingen existerande **lease**

# DISCOVER

Client → Server

- 70:5A:0F:38:C0:F9
- 00:08:E3:FF:FD:90
- 192.168.1.5

Ethernet Header — Src: **70:5A:0F:38:C0:F9** Dst: **FF:FF:FF:FF:FF:FF**

IP Header — Src: **0.0.0.0** Dst: **255.255.255.255**

UDP Header — Src: **68** Dst: **67**

**DHCPDISCOVER**

**CIADDR: 0.0.0.0**
**YIADDR: 0.0.0.0**
**SIADDR: 0.0.0.0**
**GIADDR: 0.0.0.0**
**Options: *desired options* (optional)**

# OFFER

Server → Client

- 00:08:E3:FF:FD:90
- 192.168.1.5

- 70:5A:0F:38:C0:F9

Ethernet Header — Src: **00:08:E3:FF:FD:90** Dst: **70:5A:0F:38:C0:F9**

IP Header — Src: **192.168.1.5** Dst: **0.0.0.0**

UDP Header — Src: **67** Dst: **68**

**DHCPOFFER**
**CIADDR: 0.0.0.0**
**YIADDR: 192.168.1.10**
**SIADDR: 192.168.1.5**
**GIADDR: 0.0.0.0**
**Options: *offered options***

# REQUEST

Client → Server

- 70:5A:0F:38:C0:F9
- 00:08:E3:FF:FD:90
- 192.168.1.5

Ethernet Header — Src: **70:5A:0F:38:C0:F9** Dst: **FF:FF:FF:FF:FF:FF**

IP Header — Src: **0.0.0.0** Dst: **255.255.255.255**

UDP Header — Src: **68** Dst: **67**

**DHCPREQUEST**

**CIADDR: 0.0.0.0**
**YIADDR: 192.168.1.10**
**SIADDR: 192.168.1.5**
**GIADDR: 0.0.0.0**
**Options: *requested options***

# ACK

Server → Client

- 00:08:E3:FF:FD:90
- 192.168.1.5

- 70:5A:0F:38:C0:F9

Ethernet Header — Src: **00:08:E3:FF:FD:90** Dst: **70:5A:0F:38:C0:F9**

IP Header — Src: **192.168.1.5** Dst: **0.0.0.0**

UDP Header — Src: **67** Dst: **68**

**DHCPACK**
**CIADDR: 0.0.0.0**
**YIADDR: 192.168.1.10**
**SIADDR: 192.168.1.5**
**GIADDR: 0.0.0.0**
**Options: *acknowledged options***

# IP Address Conflict?

*Kommer det bli en IP-konflikt om DHCP-servern delar ut en adress som redan används på en annan enhet?*

”As a consistency check, the allocating server SHOULD probe the reused address before allocating the address, e.g., with an ICMP echo request, and the client SHOULD probe the newly received address, e.g., with ARP.”

RFC 2131

# IP Address Conflict Avoidance

| | |
|---|---|
| 1. DISCOVER | Client → Server |
| ICMP ECHO Request | Server → ???? |
| 2. OFFER | Server → Client |
| 3. REQUEST | Client → Server |
| 4. ACK | Server → Client |
| ARP Probe | Client → ???? |

# Livelabb: Wireshark

DHCP,ICMP & ARP
ipconfig /release & /renew

# DHCP Relay

# DHCP DORA-Process

Både **DISCOVER** & **REQUEST** skickas med broadcast

→ Når enbart lokala enheter
Inom samma **brodcast domain**

Lösning: **DHCP relay**

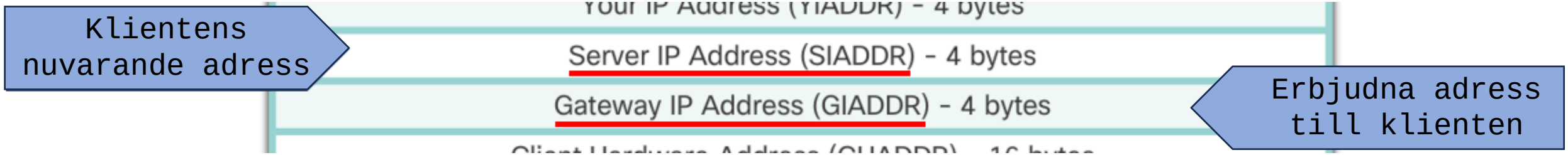
# DHCP Relay

Kallas för ’helper’ i Cisco IOS

När en DHCP broadcast träffar en **relay** skickas meddelandet vidare med **unicast**

**GIADDR**: **Relay** IP adress

**SIADDR**: DHCP server



# DHCP Redundancy

# DHCPREQUEST

*Varför skickas DHCPREQUST som broadcast?*

# Enkel redundans

Nätverket: 192.168.1.0/24

2x DHCP-servrar:

- DHCP-1 **scope**: 192.168.1.5-125
- DHCP-2 **scope**: 192.168.1.126-250

Inget överlapp!

Behöver inte synkronisera **lease** databasen

# Ordentlig redundans

- Synkronisera **lease** databasen mellan DHCP-servrar
- Görs på olika vis beroende på DHCP-servermjukvara

# DHCP Leases

# DHCP Exhaustion

**DHCP scope** töms, inga lediga **leases**

- Enkel attack:
  Spoofa MAC-adress, **DORA**-process
  ↑ repetera tills alla **leases** är tagna
- Vanlig felkonfiguration:
  För lång **lease time**

*Hur lång bör **lease time** vara?*

# Host Reservation

Ser till att en viss enhet alltid får samma IP-adress

Baserat på en option i DHCPDISCOVER

- Ofta klientens MAC-adress

# Lease Affinity

Ser till att klienter får samma adress när de förnyar sin **lease**

**DHCP-servern** väntar med att dela ut en adress som precis **expired**

**DHCP-servern** väntar i x sekunder på att klienten ska göra en ny **DORA-process**

# Lease Renewal/Rebinding

Klienten försöker förlänga sin nuvarande **lease** innan den **expire**

- Först: **Renewal T1**
- Om **T1** misslyckas: **Rebind T2**

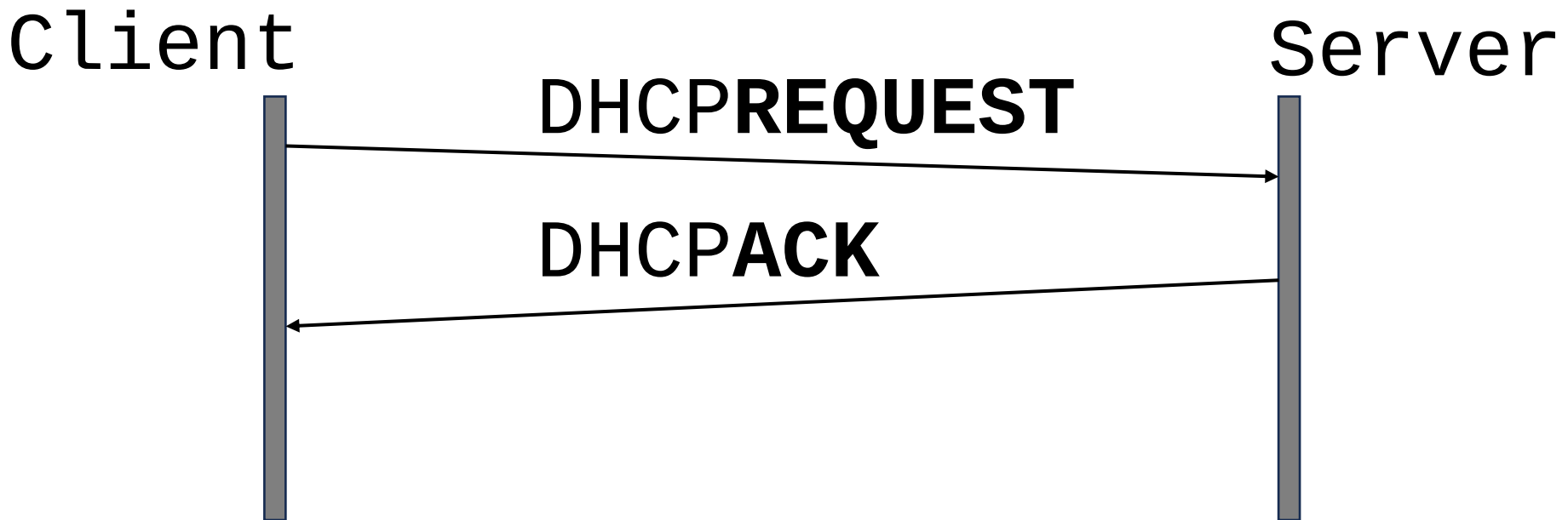
Implementeras som **options**:

| Tag | Name | Meaning |
|---|---|---|
| 58 | Renewal time | T1 |
| 59 | Rebind time | T2 |

# Renewal

**T1 renewal timer**

- Default: 50% of lease time
- Unicast REQUEST



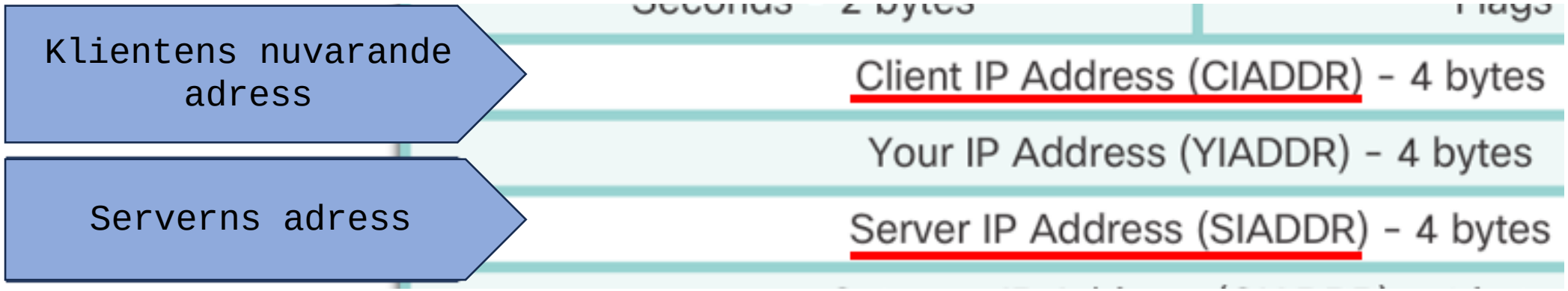
# Renewal Example

Client

- MAC address: 70:5A:0F:38:C0:F9
- CIADDR: 192.168.1.10

Server

- MAC address: 00:08:E3:FF:FD:90
- IP address: 192.168.1.5



# Renewal Example

Renewal REQUEST

Client → Server

Ethernet Header — Src: **70:5A:0F:38:C0:F9** Dst: **00:08:E3:FF:FD:90**

IP Header — Src: **0.0.0.0** Dst: **192.168.1.5**

UDP Header — Src: **68** Dst: **67**

**DHCPREQUEST**

**CIADDR: 192.168.1.10**

**YIADDR: 0.0.0.0**

**SIADDR: 192.168.1.5**

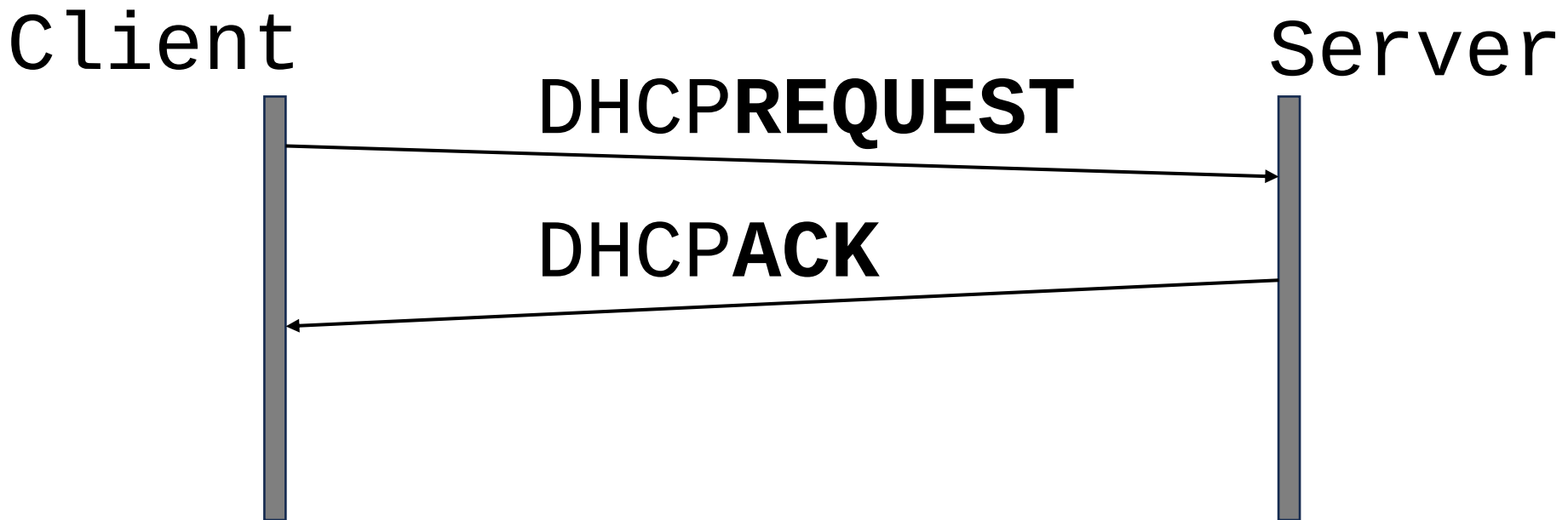
**GIADDR: 0.0.0.0**

**Options: *desired options* (optional)**

# Rebinding

**T2 rebind timer**

- Default: 87,5% of lease time
- Broadcast REQUEST



# Rebinding Example

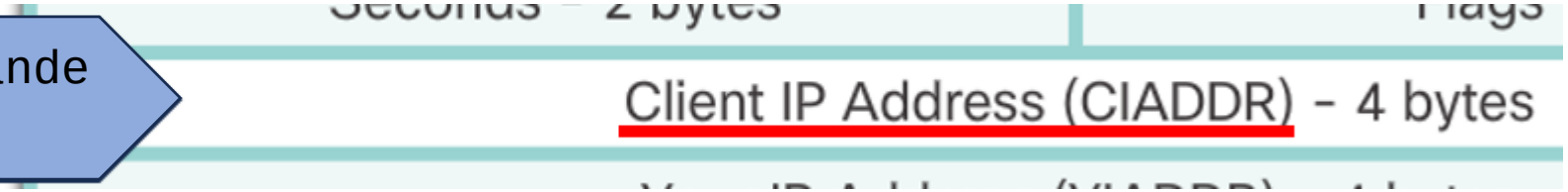
Client

- MAC address: 70:5A:0F:38:C0:F9
- CIADDR: 192.168.1.10

Server

- MAC address: 00:08:E3:FF:FD:90
- IP address: 192.168.1.5

Klientens nuvarande adress



# Rebinding Example

Rebind REQUEST

Client → Server

Ethernet Header — Src: **70:5A:0F:38:C0:F9** Dst: **FF:FF:FF:FF:FF:FF**

IP Header — Src: **0.0.0.0** Dst: **255.255.255.255**

UDP Header — Src: **68** Dst: **67**

**DHCPREQUEST**

**CIADDR: 192.168.1.10**

**YIADDR: 0.0.0.0**

**SIADDR: 0.0.0.0**

**GIADDR: 0.0.0.0**

**Options: *desired options* (optional)**

# Lease Expiry

Om både renew & rebind misslyckas

*Vad händer då om leasen expires?*

- Ny DORA-process!
- Precis som i vid första anslutning

# Sleep/hibernate

*Efter klienten kommer tillbaka från sleep/hibernate hur vet den:*

- *Om den är ansluten till samma nätverk?*
- *Om leasen tiden har passerat?*

# DHCP Options

# DHCP Options

## Definieras utav IANA

*Logo of Internet Assigned Numbers Authority*, by Internet Assigned Numbers Authority, 2015, Wikipedia (https://en.wikipedia.org/wiki/File:Internet_Assigned_Numbers_Authority_logo.svg). Fair use/I enlighet med citaträtt, copyright of IANA

https://www.iana.org/assignments/bootp-dhcp-parameters/bootp-dhcp-parameters.xhtml

### BOOTP Vendor Extensions and DHCP Options

**Registration Procedure(s)**

IETF Review

**Reference**

[RFC2939]

**Note**

The Bootstrap Protocol (BOOTP) [RFC951] describes an IP/UDP bootstrap protocol (BOOTP) which allows a diskless client machine to discover its own IP address, the address of a server host, and the name of a file to be loaded into memory and executed. The Dynamic Host Configuration Protocol (DHCP) [RFC2131] provides a framework for automatic configuration of IP hosts. The document "DHCP Options and BOOTP Vendor Information Extensions" [RFC2132] describes options for DHCP, some of which can also be used with BOOTP. Additional DHCP options are described in other RFCs, as documented in this registry.

**Available Formats**

CSV

| Tag | Name | Data Length | Meaning | Reference |
|---|---|---|---|---|
| 0 | Pad | 0 | None | [RFC2132] |
| 1 | Subnet Mask | 4 | Subnet Mask Value | [RFC2132] |
| 2 | Time Offset | 4 | Time Offset in Seconds from UTC (note: deprecated by 100 and 101) | [RFC2132] |
| 3 | Router | N | N/4 Router addresses | [RFC2132] |
| 4 | Time Server | N | N/4 Timeserver addresses | [RFC2132] |
| 5 | Name Server | N | N/4 IEN-116 Server addresses | [RFC2132] |
| 6 | Domain Server | N | N/4 DNS Server addresses | [RFC2132] |
| 7 | Log Server | N | N/4 Logging Server addresses | [RFC2132] |
| 8 | Quotes Server | N | N/4 Quotes Server addresses | [RFC2132] |
| 9 | LPR Server | N | N/4 Printer Server addresses | [RFC2132] |
| 10 | Impress Server | N | N/4 Impress Server addresses | [RFC2132] |
| 11 | RLP Server | N | N/4 RLP Server addresses | [RFC2132] |
| 12 | Hostname | N | Hostname string | [RFC2132] |

# Några vanliga options

| Tag | Name | Meaning |
|---|---|---|
| 1 | Subnet mask | |
| 3 | Router | Default gateway |
| 6 | Domain server | DNS servers |
| 12 | Hostname | |
| 15 | Domain name | |
| 43 | NTP servers | |

# **P**reboot e**X**ecution **E**nvironment

**PXE** använder **DHCP options** för att starta upp en *bootable file* över nätverket

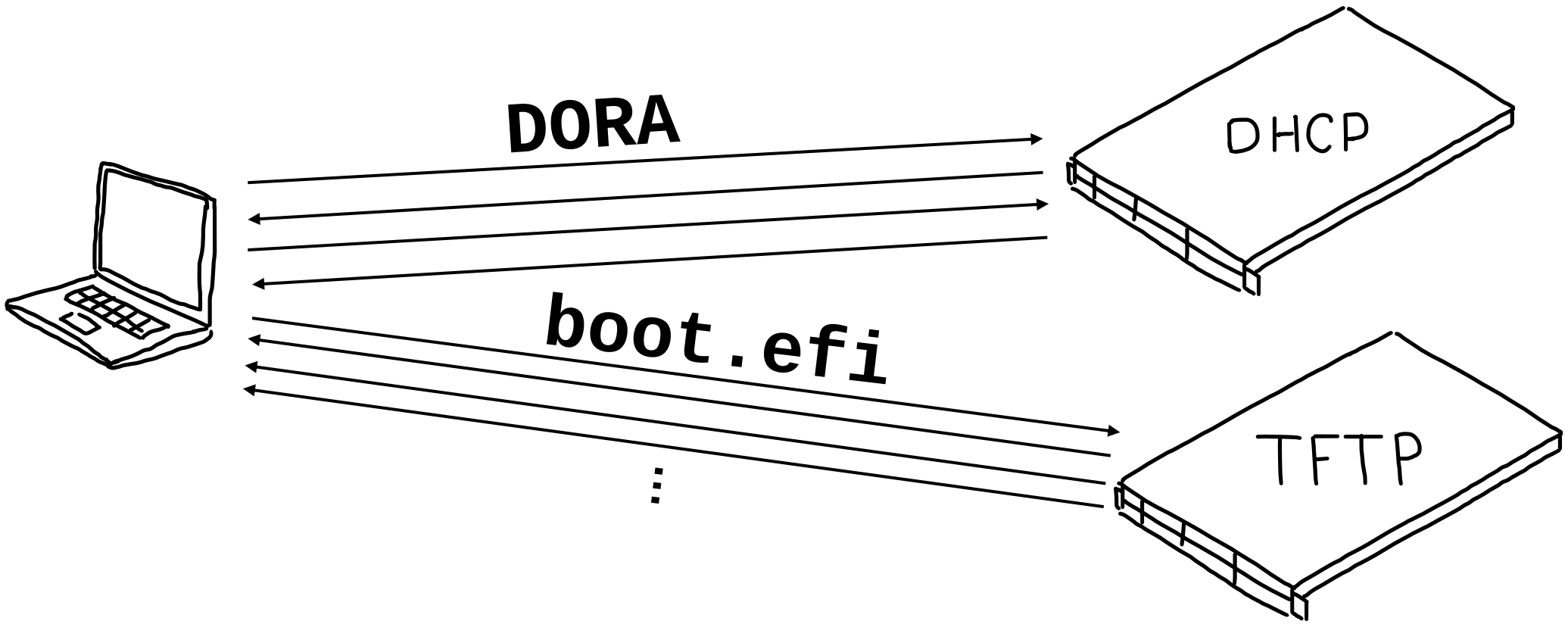
| Tag | Name | Meaning |
|---|---|---|
| 66 | Server-name | TFTP Server |
| 67 | Bootfile-name | |

Ex.

```
{
   "name": "tftp-server-name",
   "code": 66,
   "data": "10.0.250.35"
}
```

```
{
   "name": "boot-file-name",
   "code": 67,
   "data": "ipxe.efi"
}
```

# **P**reboot e**X**ecution **E**nvironment



# ISC Kea

# **I**nternet **S**ystem **C**onsortium

Utvecklar open source referens implementation baserat på RFCer

Bland annat: DHCP

# ISC DHCP

dhcpd

https://www.isc.org/blogs/isc-dhcp-eol/

Kea
Modern DHCP

Kea is an open source software system including DHCPv4, DHCPv6 servers, Dynamic DNS daemon, REST API interface, MySQL, and PostgreSQL databases, RADIUS and NETCONF interfaces and related utilities.

https://kea.isc.org/

# Livelabb: ISC Kea

Internet Systems Consortium

# DHCP Troubleshooting

# Tips

- Kan klienten nå DHCP-servern med en statisk adress (t.ex. ping)?
- Placera klienten i samma nätverk som DHCP-servern
  - Om det fungerar är det antagligen fel på din DHCP relay
- Kontrollera lease databasen! Ser du klientens MAC-adress?

169.254.x.x
What is my purpose?
You tell me there's a DHCP problem
OH MY GOD

# **A**utomatic **P**rivate **IP A**ddressing

**APIPA**: IPv4 link-local address range
169.254.0.0/16

# Till nästa gång

- Läs kurslitteraturen!
- Gör klart labb2!
- Onsdag/torsdag: labb

# Tack för idag